Q.20 Explain the working of solar pond and write its applications. (CO4)

Q.21 Describe the working principle of solar furnace. (CO4)

Q.22 Write a short note on chemical & thermochemical energy storage. (CO3)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Explain the working principle, advantages, and limitations of a photovoltaic cell. (CO1)

Q.24 Describe the working and principle of solar water pumping with suitable digram. (CO4)

Q.25 Explain the working details of natural circulation type and forced circulation type solar water heater. (CO2)

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Roll No. ....

4th Sem / Agri. Engg.

Subject : Solar Technology

Time : 3 Hrs. M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Which instrument measures direct solar radiation? (CO1)

a) Pyrheliometer b) Pyranometer

c) Sunshine Recorder d) Thermometer

Q.2 Which of the following is a commonly used battery type for solar energy storage? (CO3)

a) Alkaline b) Nickel-Cadmium

c) Lithium-ion d) Lead-carbon

Q.3 In natural circulation solar water heater, circulation of water is due to: (CO4)

a) Electricity b) Pump

c) Density difference d) None

Q.4 The current-voltage characteristics are studied for: (CO1)

a) Solar furnace b) Solar cell

c) Solar water heater d) Solar pond

Q.5 Which is not a method of solar energy storage (CO2)

a) Thermal? b) Electrical

c) Nuclear d) Chemical

Q.6 Solar pond is an example of: (CO2)

a) Thermal storage b) Chemical storage

c) Electrical storage d) None

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 The average solar constant is ________ W/m^2.(CO1)

Q.8 A photovoltaic cell converts ____________ energy into electricity. (CO1)

Q.9 Solar furnace works on the principle of _____ collectors. (CO2)

Q.10 Solar collectors are classified as ___________ and _____________ types. (CO2)

Q.11 __________ is used to dry crops using solar energy. (CO4)

Q.12 Why the inside of the box-type solar cooker is painted black? (CO4)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 State the principle of conversion of solar radiation into heat with example. (CO1)

Q.14 Explain the current-voltage characteristics of a solar cell. (CO1)

Q.15 Write short notes on Pyrheliometer and Pyranometer. (CO2)

Q.16 Classify solar collectors. (CO2)

Q.17 Describe the principle and applications of SPV module. (CO4)

Q.18 Explain advantages and limitations of concentrating collectors. (CO2)

Q.19 Define solar spectral. Also write significance of solar energy. (CO3)